

OOP-9

Total questions: 10

Worksheet time: 6mins

1.

```
int main() {  
    vector<int> v(10, 0);  
    v.push_back(3);  
    v.push_back(7);  
    println!("{}", v.size());  
}
```

a) 2

b) 10

c) 12

d) Няма да се компилира!

2.

```
class A{  
public:  
    int age;  
    A() : age(0) {}  
};  
  
A& createNewObj() {  
    A newObj;  
    return newObj;  
}  
  
int main() {  
    A obj = createNewObj();  
    println!("{}", obj.age);  
}
```

a) 0

b) 1

c) Няма да се компилира!

3.

```
class Example{
    static int count;
public:
    static void increment() {
        count++;
    }
    int getCount() {
        return count;
    }
};
int Example::count = 0;

int main() {
    Example e1;
    e1.increment();
    Example e2;
    e2.increment();
    Example::increment();
    println("{} ", e1.getCount());
}
```

a) 1

b) 3

c) Няма да се компилира!

4.

```
class Example{
public:
    int a;
    static int b;

    Example(int _a) : a(_a) {}
    void increment() {
        b++;
    }
    static void clearA() {
        a = 0;
    }
};
int Example::b = 2;

int main() {
    Example e(4);
    e.clearA();
    e.increment();
    e.increment();
    println("{} ", e.b);
}
```

a) 4

b) 2

c) Няма да се компилира!

5.

```

class A {
    int data;
public:
    A() {cout << "A() ";}
    A(int _data) : data(_data) {cout << "A(data) ";}
    A(const A& other) {cout << "A(other&) ";}
    A(A&& other) {cout << "A(other&&) ";}
    ~A() {cout << "~A() ";}
};

class B {
    A a;
};

int main() {
    B b1;
    B b2 = b1;
}

```

a) A() A(other&) ~A() ~A()

b) A() ~A()

c) Няма да се компилира!

6.

```

class Counter {
public:
    int count = 0;
    Counter(int _count = 0) : count(_count) {}
    void operator+=(const Counter& other) {
        count += other.count;
    }
};

int main() {
    Counter c1(3);
    Counter c2 = c1;
    c2 += c1;
    cout << c2.count << endl;
}

```

a) 3

b) 6

c) Няма да се компилира!

7.

```

class A {
    friend class B;
    int data;
public:
    A(int _data) : data(_data) {cout << "Constructor ";}
    A(const A& obj) : data(obj.data) {cout << "Copy constructor ";}
    ~A() {cout << "Destructor ";}
};

class B {
    int data;
public:
    B(int _data) : data(_data) {}
    bool isEqualToA(A a) {
        if (data == a.data)
            return true;
        return false;
    }
};

int main() {
    A a(1);
    B b(1);
    cout << b.isEqualToA(a) << " ";
}

```

a) Constructor Copy constructor 1 Destructor
Destructor

b) Constructor Destructor

c) Няма да се компилира!

8.

```
class Example {
public:
    int data = 2;

    Example& operator*=(int scalar) {
        data *= scalar;
        return *this;
    }
    Example& operator*=(const Example& other) {
        data *= other.data;
        return *this;
    }
};

int main() {
    Example e1;
    e1 *= 2;
    cout << e1.data << " ";
    Example e2;
    e1 *= e2;
    cout << e1.data;
}
```

a) 4 8

b) 2 2

c) Няма да се компилира!

9.

```
class A {
public:
    int data = 0;
};

class B {
    A a;
public:
    A& getA() {
        return a;
    }
};

int main() {
    B b;
    b.getA().data = 67;
    cout << b.getA().data;
}
```

a) 0

b) 67

c) Няма да се компилира!

10.

```
class Modifier {
    Modifier() {}
public:
    Modifier(const Modifier& other) = delete;
    Modifier& operator=(const Modifier& other) = delete;

    static Modifier& getInstance() {
        static Modifier instance;
        return instance;
    }

    void changeVector(vector<int> vec, int ind, int val) {
        vec[ind] = val;
    }
};

int main() {
    Modifier& mod = Modifier::getInstance();
    vector<int> v(3, 0);
    mod.changeVector(v, 0, 1);
    mod.changeVector(v, 1, 2);

    for (int i = 0; i < 3; i++)
        cout << v[i] << " ";
}
```

a) 0 0 0

b) 1 2 0

c) Няма да се компилира!

Answer Keys

1. c) 12
2. c) Няма да се компилира!
3. b) 3
4. c) Няма да се компилира!
5. a) A() A(other&) ~A() ~A()
6. b) 6
7. a) Constructor Copy
constructor 1 Destructor
Destructor
8. a) 4 8
9. b) 67
10. a) 0 0 0

